

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12400194
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Jamkhedkar; Prashant et al.

Unified transaction services for multi-tenant architectures

Abstract

A method for using unified transaction services in a multi-tenant architecture system is discussed. The method includes receiving a request, at a first service provider, to provide a first transaction service for a user. The method includes accessing a first representation of the first service provider in a first hierarchical data structure, the first hierarchical data structure being managed by a second service provider, the second service provider managing user identity of the user. The method includes determining, based on the first representation, that transaction resources required for completion of the first transaction service are provided at the second service provider using a resource representation. The method also includes, responsive to determining that the transaction resources are accessible at the first service provider, accessing, at the first service provider, the transaction resources via the resource representation.

Inventors: Jamkhedkar; Prashant (Fremont, CA), Ranganathan; Aravindan (San Jose, CA), Kumar; Sandeep (Fremont, CA), Aoki; Norihiro (Woodside, CA), White; Justin (La Selva, CA), Meyer; Jeffrey (Los Gatos, CA)

Applicant: PAYPAL, INC. (San Jose, CA)

Family ID: 1000008782797

Assignee: PAYPAL, INC. (San Jose, CA)

Appl. No.: 17/462750

Filed: August 31, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20210398089 A1	Dec. 23, 2021

Related U.S. Application Data

Publication Classification

Int. Cl.: **G06Q20/02** (20120101); **G06F16/22** (20190101); **G06F16/23** (20190101); **G06Q20/08** (20120101)

U.S. Cl.:

CPC **G06Q20/027** (20130101); **G06F16/2246** (20190101); **G06F16/2379** (20190101);
G06Q20/08 (20130101);

Field of Classification Search

CPC: G06Q (20/027); G06Q (20/08); G06F (16/2379); G06F (16/2246)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7657639	12/2009	Hinton	N/A	N/A
8271536	12/2011	Amradkar et al.	N/A	N/A
8606704	12/2012	Roberts	705/40	G07F 17/0014
8819801	12/2013	Sangubhatla et al.	N/A	N/A
8918091	12/2013	Hoelzle et al.	N/A	N/A
8918641	12/2013	Smith et al.	N/A	N/A
9058481	12/2014	Prasad et al.	N/A	N/A
9218599	12/2014	Hilbring	N/A	N/A
9710664	12/2016	Sathyadevan et al.	N/A	N/A
9781122	12/2016	Wilson	N/A	N/A
9992186	12/2017	Drozd et al.	N/A	N/A
10218703	12/2018	Beiter	N/A	N/A
10581867	12/2019	Srinivasan	N/A	H04L 63/104
10853763	12/2019	Armstrong et al.	N/A	N/A
10931656	12/2020	Carru et al.	N/A	N/A
11734658	12/2022	Jamkhedkar et al.	N/A	N/A
2002/0010591	12/2001	Pomerance	N/A	N/A
2002/0133412	12/2001	Oliver	705/26.8	G06Q 30/02
2004/0117298	12/2003	Algiene et al.	N/A	N/A
2004/0128392	12/2003	Blakley et al.	N/A	N/A
2005/0261968	12/2004	Randall et al.	N/A	N/A
2006/0059565	12/2005	Green et al.	N/A	N/A
2007/0150480	12/2006	Hwang et al.	N/A	N/A
2007/0294405	12/2006	Mohindra	709/226	H04L 67/51

2008/0154783	12/2007	Rule et al.	N/A	N/A
2008/0288400	12/2007	Panthaki et al.	N/A	N/A
2009/0125972	12/2008	Hinton et al.	N/A	N/A
2011/0276674	12/2010	Jensen-Horne et al.	N/A	N/A
2011/0276892	12/2010	Jensen-Horne et al.	N/A	N/A
2012/0072307	12/2011	Kassaei et al.	N/A	N/A
2012/0197963	12/2011	Bouw et al.	N/A	N/A
2012/0218917	12/2011	Komarevtsen	N/A	N/A
2012/0258777	12/2011	Huang	N/A	N/A
2013/0046600	12/2012	Coppinger	N/A	N/A
2013/0111558	12/2012	Sangubhatla et al.	N/A	N/A
2013/0132404	12/2012	Liu	707/748	G06F 16/27
2013/0318241	12/2012	Acharya et al.	N/A	N/A
2014/0230076	12/2013	Micucci et al.	N/A	N/A
2015/0161620	12/2014	Christner	N/A	N/A
2015/0193775	12/2014	Douglas et al.	N/A	N/A
2015/0193858	12/2014	Reed et al.	N/A	N/A
2015/0348003	12/2014	Reader	N/A	N/A
2016/0087960	12/2015	Pleau et al.	N/A	N/A
2016/0092859	12/2015	Klingen	N/A	N/A
2016/0142409	12/2015	Frei	713/176	G06F 21/33
2016/0203538	12/2015	Smits et al.	N/A	N/A
2016/0292694	12/2015	Goldschlag et al.	N/A	N/A
2016/0314460	12/2015	Subramanian	N/A	G06Q 20/4014
2016/0335118	12/2015	Beiter et al.	N/A	N/A
2016/0337365	12/2015	Beiter	N/A	H04L 63/101
2017/0012962	12/2016	Lissack et al.	N/A	N/A
2017/0048306	12/2016	Amador	N/A	N/A
2017/0318128	12/2016	Ananthanarayanan et al.	N/A	N/A
2017/0323117	12/2016	Pleau et al.	N/A	N/A
2018/0013763	12/2017	Wilson et al.	N/A	N/A
2018/0075231	12/2017	Subramanian et al.	N/A	N/A
2018/0081905	12/2017	Kamath et al.	N/A	N/A
2018/0309759	12/2017	Leibmann et al.	N/A	N/A
2019/0014120	12/2018	Drabant	N/A	N/A
2019/0026486	12/2018	Pleau	N/A	H04L 63/104
2019/0097802	12/2018	Rowe et al.	N/A	N/A
2020/0007529	12/2019	Bahrenburg et al.	N/A	N/A
2020/0412538	12/2019	Rosado	N/A	H04L 63/126
2021/0136083	12/2020	Gordon et al.	N/A	N/A
2021/0226788	12/2020	Bahrenburg et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
------------	------------------	---------	-----

105830388	12/2015	CN	N/A
105917309	12/2015	CN	N/A
107852417	12/2017	CN	N/A
108322472	12/2017	CN	N/A
108337260	12/2017	CN	N/A
3361700	12/2020	EP	N/A

OTHER PUBLICATIONS

Extended European Search Report for European Application No. 19903189.9 mailed on Jan. 21, 2022, 10 pages. cited by applicant

Ashwini A., "What Are the Pros and Cons of Multi-tenancy in Rails Application?", Retrieved from the Internet: <https://medium.com/swlh/what-are-the-pros-and-cons-of-multi-tenancy-in-rails-application-547d0f9c205f>, Oct. 3, 2017, 9 pages. cited by applicant

Axero., "Single Tenant vs Multi Tenant Business Software", Communifire, Retrieved on Apr. 30, 2018, Retrieved from the Internet URL: <https://axerosolutions.com/blog/single-tenant-vs-multi-tenant-business-software>, 9 pages. cited by applicant

Crown N., "An Introduction to SCIM: System for Cross-Domain System for Cross-Domain Identity Management", RSAConference, Europe, 2012, Retrieved on Apr. 24, 2018, Retrieved from the Internet URL: <https://docs.huihoo.com/rsaconference/europe-2012/How-SCIM-Simplifies-Migration-to-Cloud-Services.pdf>, 24 pages. cited by applicant

Ed K.L., et al., "System for Cross-Domain Identity Management: Definitions, Overview, Concepts and Requirements" Retrieved on Apr. 24, 2019, Retrieved from the Internet URL: <https://www.rfc-editor.org/rfc/pdf/rfc7642.txt.pdf>, 19 pages. cited by applicant

European Application No. 19819808.7, Extended European Search Report mailed on Jun. 8, 2021, 7 pages. cited by applicant

International Appl. No. PCT/US2019/037242, International Preliminary Report on Patentability mailed on Dec. 24, 2020, 12 pages. cited by applicant

International Appl. No. PCT/US2019/037242, International Search Report and Written Opinion mailed on Oct. 4, 2019, 12 pages. cited by applicant

International Appl. No. PCT/US2019/064325, International Search Report and Written Opinion mailed Mar. 24, 2020, 16 pages. cited by applicant

International Application No. PCT/US2019/064325, International Preliminary Report on Patentability mailed on Jun. 17, 2021, 6 pages. cited by applicant

International Application No. PCT/US2019/068993, International Preliminary Report on Patentability mailed on Jul. 8, 2021, 8 pages. cited by applicant

International Application No. PCT/US2019/068999, International Preliminary Report on Patentability mailed on Jul. 8, 2021, 7 pages. cited by applicant

International Application No. PCT/US2019/068999, International Search Report and Written Opinion mailed on Mar. 31, 2020, 8 pages. cited by applicant

International Search Report and Written Opinion for International Application No. PCT/US2019/68993 mailed on Mar. 31, 2020, 14 pages. cited by applicant

International Search Report and Written Opinion for International Application No. PCT/US2019/68999 mailed on Mar. 31, 2020, 14 pages. cited by applicant

Jones M., et al., "JSON Web Token (JWT)," Internet Engineering Task Force (IETF), May 2015, 30 pages. cited by applicant

Kabbedijk J., et al., "Multi-Tenant Architecture Comparison", Retrieved on Apr. 30, 2018, Retrieved from the Internet URL:

<https://pdfs.semanticscholar.org/17fb/e778244aa190da9d56c5fb8c2e64c63cd621.pdf>, 8 pages. cited by applicant

Microsoft., "Chapter 2: Key Principles of Software Architecture," Retrieved on Apr. 30, 2018, 6

pages. cited by applicant
Mundra M., “Multi-Tenant Vs. Single-Tenant Architecture (SaaS),” Retrieved from the Internet URL: <https://blogs.sap.com/2015/07/12/multi-tenant-vs-single-tenant-architecture-saas/>, Jul. 12, 2015, 5 pages. cited by applicant
Nidhiben S., et al., “Multi-Tenant Access and Information Flow Control for SaaS”, IEEE International Conference on Web Services, Jun. 27, 2016, Retrieved from the Internet URL: <https://ieeexplore.ieee.org/document/7557990>, Retrieved on Mar. 3, 2021, 8 pages. cited by applicant
Sageintacct., “Multi-Tenancy is Fundamental to SaaS Applications—and It's Better for You, Too” Retrieved on Dec. 7, 2018, Retrieved from the Internet URL: <https://www.sageintacct.com/cloud-erp/multi-tenant-cloud-architecture>, 4 pages. cited by applicant
Multitenancy—Wikipedia; Oct. 18, 2019; <https://en.wikipedia.org/w/index.php?title=Multitenancy&oldid=598245774>; pp. 1-5. cited by applicant
First Office Action for Chinese Application No. 201980052687.4, mailed on Jan. 30, 2024, 13 pages. cited by applicant
First Office Action for Chinese Application No. 201980087063.6, mailed on Dec. 28, 2023, 5 pages. cited by applicant
Mohamed F., et al., “An Integrated Platform for Dynamic Adaptation of Multi-Tenant Single Instance SaaS Applications,” 2017 IEEE 5th International Conference on Future Internet of Things and Cloud, 2017, pp. 257-264. cited by applicant
Zheng J., “A Framework for Migration of Conventional Client-Server Software Systems to Cloud,” May 2014, The University of New Brunswick, 114 pages. cited by applicant
European Patent Application No. 19819808.7, European Patent Office, Examination search report dated Dec. 23, 2024, 8 pages. cited by applicant

Primary Examiner: Nolan; Peter D

Assistant Examiner: Cobb; Matthew

Attorney, Agent or Firm: HAYNES AND BOONE, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of and is a continuation of U.S. application Ser. No. 16/213,811, filed Dec. 7, 2018, which is a continuation-in-part of U.S. application Ser. No. 16/009,994, titled “Unified Identity Services for Multi-Tenant Architectures,” filed on Jun. 15, 2018 and issued as U.S. Pat. No. 11,030,329 on Jun. 8, 2021, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

(1) Embodiments of the present disclosure generally relate to the field of software architecture and, more particularly, to managing how various entities are on-boarded, managed, and/or accessed in a multi-tenant system architecture.

(2) Complex software systems can have various architectures, including a single-tenant software architecture and a multi-tenant software architecture, among others. In a single-tenant software architecture, each entity (such as a company or a portion of that company) can have their own instances of each software application, data, and any supporting infrastructure. In a single-tenant architecture, the software for each tenant entity can be customized as desired. However, drawbacks of using the single tenant architecture include expense in resources, as well as requirements to host,

customize, and maintain separate software applications for each tenant.

(3) In contrast, in a multi-tenant software architecture, each entity can share the same instances of applications, data, and/or infrastructure. A multi-tenant software provider can provision their applications and/or resources to multiple entities from a single software architecture. In multi-tenant software architecture, access to data can be shared among the various entities. By sharing much of the applications, data, and software, various resources such as installation, configuration, number of physical servers, maintenance, and even power can be optimized. However, drawbacks of using multi-tenant architecture include complexity of managing how multiple software applications are configured and shared among multiple clients. Also, design of multi-tenant architecture-based software systems that provides secure and reliable access to the data, resources, and/or transaction services that operate using the data and/or resources can be problematic. Furthermore, interoperability difficulties may arise when using software applications, in a multi-tenant architecture system, that were initially managed and/or hosted by different businesses.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present embodiments may be better understood, and numerous objects, features, and advantages made apparent to those skilled in the art by referencing the accompanying drawings.

(2) FIG. 1 is a system diagram illustrating embodiments of a multi-tenant software architecture system that can be accessed by communicating with a user device.

(3) FIG. 2 is a system diagram illustrating embodiments of the multi-tenant software architecture of FIG. 1 including unified transaction services.

(4) FIG. 3 is a diagram illustrating embodiments of flow of operations of using the multi-tenant software architecture for unified transaction services.

(5) FIG. 4 is a diagram illustrating hierarchical data structures used by the multi-tenant software architecture systems that provide unified transaction services.

(6) FIG. 5 is a diagram illustrating embodiments of relationships between a multi-tenant identity service, multi-tenant transaction resource service, and multi-tenant social graph service for unified transaction services.

(7) FIG. 6 is a flow diagram illustrating embodiments of operations for using multi-tenant software architecture systems that provide unified transaction services.

(8) FIG. 7 is a block diagram illustrating embodiments of electronic devices used in the communication systems of FIGS. 1-6.

DESCRIPTION OF EMBODIMENT(S)

(9) The description that follows includes exemplary systems, methods, techniques, instruction sequences, and/or computer program products that embody techniques of the present disclosure. However, it is understood that the described embodiments may be practiced without these specific details. For instance, although some examples refer to accessing tenant services, other types of service providers are contemplated, such as of Software-as-a-Service (SaaS), among others.

(10) In a multi-tenant software architecture described herein, each tenant can share the applications, data, and/or infrastructure. A multi-tenant software provider can use a multi-tenant software platform to provision applications and/or resources to multiple entities. The multi-tenant software platform can facilitate addition of new tenants and onboard data and/or services provided by these new tenants. The multi-tenant software platform can implement rules and policies for data access by various entities across the tenants. The multi-tenant software platform can use identity services to provide access to these services, such as from an entity associated one tenant to a service provided by another tenant. The multi-tenant software platform can facilitate transaction services between tenants, such as a transaction service originating at one tenant that accesses resources at

another tenant.

(11) The multi-tenant software platform (also referred to as a multi-tenant platform) can include multiple tenants that have been previously on-boarded. The multi-tenant platform can facilitate access, using a unified identity associated with a user, to the entities, policies, and services of these tenants. The tenants can be hosted and managed by service providers. Access to the services can be determined based on privileges of the unified identity, as well as on policies of each of the tenants. The services can include various merchant services, such as in-store check-out (at a certain tenant) via a user device (associated with the unified identity), access to an online store of the tenant, order-ahead at a certain store of the tenant, a cash-in process (e.g., at an Automated Teller Machine (ATM)) at a certain tenant, a cash-out process (e.g., at an ATM), self-checkout for fuel at pay-at-the-pump stations of the tenant, among others.

(12) The services can include SaaS and Platform as a Service (PaaS) services and/or other cloud services that are accessible by the user device. For example, the solution provider can provide access, to the user device, to various software applications, or deliver such software to the user device. The services can also include inter-tenant transaction services, where a transaction service that is originated at a first tenant needs to access transaction resources at a second tenant. The multi-tenant platform can access other services, such as multi-tenant identity services, multi-tenant social graph service, and/or multi-tenant transaction resource service to determine how to process a certain inter-tenant transaction service.

(13) The multi-tenant platform can thus utilize unified identity services to provide access to selected services and/or data of the tenants based on policies associated with each of the tenants. The multi-tenant platform can use separate databases to store data to achieve isolation, such as when new tenants are on-boarded onto the existing platform, and to provide logical and/or physical data isolation. Shared data access can be made possible by on-boarding the tenants and selectively cross-exposing services. For example, for a multi-tenant platform managed by PAYPAL, certain capabilities of the multi-tenant platform such as Risk-as-a-Service (RaaS) and/or TRaaS (Transaction Resource as a Service) can be provided to certain users of on-boarded tenants such as XOOM. Similarly, certain capabilities of XOOM such as Fund Sending can be provided to certain users of PAYPAL. Thus, in some embodiments, the multi-tenant platform can include a core tenant (such as PAYPAL) that offers core services and infrastructure (including Identity-as-a-Service (IaaS) functionality and TRaaS) and any additional data access to selected tenants and/or users accessing the multi-tenant platform from that tenant. The level of access to the core services and/or infrastructure can be determined by rules and/or policies of each tenant and/or of the multi-tenant platform.

(14) The multi-tenant platform can onboard new tenants by using a one or more hierarchical data structures. In some embodiments, the multi-tenant platform can expose appropriate user experiences for a user application based on the user request. The multi-tenant platform can implement methods for providing IaaS services to various entities and/or tenants. The multi-tenant platform can provide the IaaS and other services based on the hierarchical data structures. As discussed below, the multi-tenant platform can process user requests (e.g., from users of the core tenant or from users of an on-boarded tenant) and determine how to access and process services, including inter-tenant services.

(15) In some embodiments, the multi-tenant platform can receive a request, at a first service provider, to provide a first transaction service for a user. The multi-tenant platform can access a first representation of the first service provider in a first hierarchical data structure that is managed by a second service provider, where the second service provider manages user identity of the user. The multi-tenant platform can determine, based on the first representation, that transaction resources required for completion of the first transaction service are provided at the second service provider using a resource representation. The multi-tenant platform can, responsive to determining that the transaction resources are accessible at the first service provider, access, at the first service

provider, the transaction resources via the resource representation. The following description and associated Figures illustrate various embodiments directed to the ideas listed above.

(16) FIG. 1 is a system diagram illustrating embodiments of a multi-tenant software architecture system that can be accessed by communicating with a user device. In an overview of the system diagram, a user device **104** can communicate with a processing system **100** that includes a multi-tenant platform **102**. The processing system **100** can process requests using a multi-tenant transaction processor (mt-transaction processor) **101**. The multi-tenant platform **102** can provide access to multiple service providers **108**, **110**, and **112**. In some embodiments, the multi-tenant platform **102** can model each of the service providers **108-112** such that the mt-transaction processor **101** can access services **116-124**, policies **123-127**, and/or resources **162-166** of the service providers.

(17) The multi-tenant platform **102** includes core services at elements **132-148**, as well as additional services **130(1)-130(5)**. The multi-tenant platform **102** can provide services to the service providers **108-112**, such as IaaS services that may be required for accessing other services. Thus, the multi-tenant platform **102** can manage a plurality of tenants, each one of which can be associated with one or more services, which are then exposed for access. In some embodiments, each of the service providers **108-112** can be the actual providers that are then represented (e.g., by modeling) in the multi-tenant software architecture (i.e., in the multi-tenant platform **102**).

Although not shown in FIG. 1, each of the service providers **108-112** can include a respective transaction processor for processing own requests. The mt-transaction processor **101** can be a transaction processor that was fully integrated onto the multi-tenant platform **102**. The mt-transaction processor **101** can be a default transaction processor for processing transactions.

(18) The multi-tenant platform **102** can store user information for users at the accounts **142**. In some embodiments, the accounts **142** includes information for users (also referred to as “core users”) of a core tenant. The core tenant can be one of the service providers **108-112**, or a service provider that is fully integrated into the multi-tenant platform **102**. The multi-tenant platform **102** can facilitate provision of various core services to the core users. The core services can include identity services **132**, risk services **138**, compliance services **140**, and payment services **148**. In one embodiment, the multi-tenant platform **102** provides IaaS services at the identity services **132** element. The identity services **132** can generate and maintain a core hierarchical data structure for managing its core entities. The core services can also include access to policy configuration **136**, as well as access to accounts **142**, merchants **144**, and consumers **146**. The accounts **142** are associated with the identity services **132**. The multi-tenant platform **102** can also provide inter-tenant transaction services, such as between tenants **108** and **110**, where the tenants **108** and **110** need not be fully integrated.

(19) The processing of a certain transaction can be initiated by the mt-transaction processor **101** by using a single instance of an MT-IDP (such as the MT-IDP **246** discussed below) to access resources (as required by the transaction) at a certain service provider. In one example, the processing system **100** can process a transaction service that is requested at the service provider **108**. For example, the mt-transaction processor **101** can use the multi-tenant platform **102** to process a request that is received from the service provider **108** for use of certain transaction resources, which can be provided by the multi-tenant platform **102** and/or by other service providers **110-112**. These transaction resources can be accessed, at the multi-tenant platform **102**, via a resource representation. The multi-tenant platform **102** can access the identity **132** services, transaction resource service **150** (referred to also as MT-FI **150**), and/or the social graph service **160** (referred to also as MT-SG **160**) to determine the resource representation.

(20) The service provider **108** can include entities **114**, policies **123**, services **116**, and transaction resources **162**. Similarly, the service providers **110** and **112** include entities **118** and **122**, policies **125** and **127**, services **120** and **124**, and/or transaction resources **164** and **166**, respectively. The elements **114**, **123**, **116**, and **162** of the service provider **108**, and similar elements of the service

providers **110** and **112** can be modeled by the multi-tenant platform **102**, as discussed below. The multi-tenant platform **102** can determine how transaction resources (for a certain transaction) are being provided, such as by redirecting the request to use actual transaction resources that are hosted by one of the other service providers **110** or **112**. For example, the actual transaction resources at the service provider **112** can be accessed by the mt-transaction processor **101** through the multi-tenant platform **102** via the resource representation. The multi-tenant platform **102** can thus facilitate processing of transactions using a unified multi-tenant architecture for transactions (i.e., service transactions) that originate at one tenant and use resources at another tenant. As discussed below, the multi-tenant platform **102** can determine to use another transaction processor (e.g., an actual transaction processor of the service provider **110**) for processing of the transaction.

(21) FIG. 1 is a system diagram illustrating embodiments of a multi-tenant software architecture system that can be accessed by communicating with a user device. In an overview of the system diagram, a user device **104** can communicate with a processing system **100** that includes a multi-tenant platform **102**. The processing system **100** can process requests using a multi-tenant transaction processor (mt-transaction processor) **101**. The multi-tenant platform **102** can provide access to multiple service providers (also referred to as tenants) **108**, **110**, and **112**. In some embodiments, the multi-tenant platform **102** can model each of the service providers (i.e., tenants) **108-112** such that the mt-transaction processor **101** can access services **116-124**, policies **123-127**, and/or resources **162-166** of the service providers.

(22) In some embodiments, the mt-transaction processor **101** can access, via the resource representation, a modeled resource as modeled by the multi-tenant platform **102**. The multi-tenant platform **102** then can facilitate access to the actual transaction resource that is associated with the modeled resource. Thus, the actual transaction resources **166** at the service provider **112** can be accessed by the mt-transaction processor **101** through the multi-tenant platform **102** via the resource representation. The actual transaction resource at the service provider **112** can be consumed. For example, the actual transaction resources **166** can correspond to a certain financial instrument that is on-boarded by the service provider **112**. The actual transaction resources **166** can correspond to incentives that are associated with the user's account. In another example, the actual transaction resource **166** can correspond to certain memory and/or computing resources (e.g., for a certain requested SaaS transaction). Upon processing, a value of that transaction resource at one tenant can be changed, such as decremented from one value to another value (and potentially incremented at another value of another transaction resource at another tenant).

(23) The multi-tenant platform **102** can onboard the service providers **108-112**, such as to generate models of the respective service provider(s). As discussed below, the onboarding process can include accessing identity services to determine at least a portion of the entities that are being managed by the respective identity service. If the entities are not being migrated (e.g., by being fully integrated) to the identity services **132**, then the identity services can generate representations of the entities being on-boarded in the core hierarchical data structure. These representations are then being used by the IaaS service to determine and use the unified identity (for a certain user) across the multi-tenant platform **102**. The representations can include lightweight elements and/or linked elements. The identity services **132** can manage and access the policy configuration **136**, such as to enforce access to certain customer and/or merchant domains. The policy configuration **136** can be accessed via the hierarchical data structures (discussed below) for each entity. The process of generating and using user representations is explained in more detail below.

(24) The multi-tenant platform **102** can communicate, via the multi-tenant API **126**, with the service providers **108-112** and/or the user device **104**. The multi-tenant platform **102** can provide, based on a certain unified identity, information related to a tenant, associated services, and/or associated user experiences, to a requesting entity via the multi-tenant tenant API **126**.

Furthermore, the multi-tenant platform **102** can facilitate communication between various tenants **108-112**, such as by providing IaaS services and use of unified identity.

(25) For example, a new tenant, such as the service provider **112**, can be on-boarded by the multi-tenant platform **102**. Information on the new tenant can be stored by one or more of the core services **132-148**. In some implementations, upon onboarding, a namespace (e.g., its entities such as merchants and/or customers) that is associated of each tenant can be onboarded onto the identity services **132**. In some embodiments, one or more of services of the new tenant, such as the service **124**, can be exposed for access to other users of the multi-tenant platform **102**. In one embodiment, the multi-tenant platform **102** can also generate a representation **130(5)** of the service **124** at the multi-tenant platform **102**. Thus, any access from other tenants, such as from the user device **104** or from the service provider **110**), to the service **124** is performed via a unified identity (e.g., by using IaaS) at the service representation **130(5)**. In some embodiments, for onboarded tenants that are fully integrated, access to services of that tenant can be performed by the mt-transaction processor **101** directly, without using representations of the fully integrated service. In some embodiments, for onboarded tenants that are not fully integrated, access to services of that tenant can be performed by the mt-transaction processor **101** by calling the representation **130(5)** of the onboarded (but not fully integrated) service. The representation **130(5)** can then access the actual service **124**.

(26) The user device **104** can be any type of a computing device that can communicate user requests to the multi-tenant platform **102**. The user device **104** can be implemented as a kiosk, a point-of-sale (PoS) terminal, a mobile device, among others. The user device **104** includes a user interface (UI) **106** through which the user can interact with displayed user experiences, such as to access certain services provided via the multi-tenant platform **102**. The user device **104** can generate and communicate a user request for a certain service at the multi-tenant platform **102**. The user device **104** can interface with the service provider **108-112** and/or with the processing system **100**. The transaction processor then can communicate with various components of the multi-tenant platform **102**, such as via the multi-tenant API **126**.

(27) In some embodiments, one of the tenants **108-112** can implement one or more payment system functions. In one embodiment, the core service provider (that is integrated into the multi-tenant platform **102**) can process a payment and order fulfilment for services provided via user experiences accessed at the UI **106**. The core service provider can perform risk analysis (e.g., via the Risk core service **138**) to determine whether to perform a certain service and/or process a payment for that service. In some embodiments, the core service provider can perform risk and/or verification services for the new tenant **112** being on-boarded by the multi-tenant platform **102**. The core service provider can process payments from the user account that is associated with the user device **104**. The core service provider can provide financial services, such as a fund transfer (e.g., a transfer of a certain monetary amount), to the user. The core service provider can include payment accounts, each of which can be associated with a buyer or a seller.

(28) Once on-boarded as a core tenant, the accounts of the core service provider can be stored and accessed directly at the multi-tenant platform at the accounts **142** element. For example, a buyer (e.g., a user of the user device **104**) can be associated with one payment account, and the seller (e.g., one of the tenants) can be associated with another payment account at the payment system (which can be implemented using the processing system **100**). Upon successfully performing the risk analysis on the requested service (e.g., a requested transaction at the tenant), the core service provider can then perform (e.g., via the payment core service **148**) a fund transfer from the buyer's payment account to the seller's payment account.

(29) The payment system can be implemented by PAYPAL or another online payment system that allows users to send, accept, and request fund transfers. In some embodiments, the user experience can also provide access to certain services. Thus, in addition, or instead of, a payment service, the user experience can include other functionality that is unique for the certain tenant, such as selection of items for order, access to certain SaaS functionality, among others. Thus, the solution provider can provide funds in/funds out services, as determined based on the location data, from

the payment system to the users of the user devices.

(30) For simplicity, FIG. 1 only shows a single user device **104**. However, as discussed herein, the multi-tenant platform **102** interfaces with multiple user devices, and thus provides access to services to many different users. Similarly, the multi-tenant platform **102** can onboard multiple tenants in addition to the ones shown. The multi-tenant platform **102** can also have multiple core tenants that are incorporated as part of the core services **132-148**.

(31) FIG. 2 is a system diagram illustrating embodiments of the multi-tenant software architecture of FIG. 1 including unified transaction services. FIG. 2 shows embodiments of how the identity services of the service providers **108** and **110** are modeled by the multi-tenant platform **102**. As shown, the (actual) service provider **108** can include a third-party identity provider (3P-IDP) **204** that manages entities of merchant **206(1)** and **206(2)**, and entities of consumer **202(1)** and **202(2)**. Similarly, the (actual) service provider **110** can include a 3P-IDP **204** that manages entities of consumer **214(1)** and **214(2)**. In some examples, prior to being on-boarded, the service provider **108** is operable to perform services **116** (optionally based on the policies **123**) for the entities **114** using the resources **162**, without using the multi-tenant platform **102**.

(32) The multi-tenant platform **102** includes the identity service **132**, the accounts **142** element, the risk **138** element, as well as services **130(1)** and **130(2)**. The multi-tenant platform **102** also includes the identity service **132**, the accounts **142** element, the risk **138** element, as well as services **130(1)** and **130(2)**. The service provider **108** can be modeled by the multi-tenant platform **102** by the identity service **132**, e.g., as part of the hierarchical data structure **244**. As discussed below with reference to FIGS. 3-5, the identity service **132** can model some parts of the service provider **108**, including the 3P-IDP **204**, the merchants **206(1)** and **206(2)**, the consumers **202(1)** and **202(2)**, and/or the FIs **209(1)** and **209(2)**. The identity service **132** can similarly model some parts of the service provider **110** and/or other service providers. The identity service **132** can model each of the service providers **108** and/or **110** as service provider representations that can be generated during onboarding of the respective service providers **108** and/or **110**. The 3P-IDPs **204** and **212** can be represented, in the hierarchical data structure **244**, of the actual IDPs provided by the service providers **108** and **110**, respectively.

(33) The identity service **132** includes an access tier **242** and the hierarchical data structure **244** that is managed by an MT-IDP **246**. The multi-tenant platform **102** also includes services **138**, **142**, and **130(1)-130(2)**. The access tier **242** is a part of the identity service **132** that customizes data associated with the service provider when being on-boarded with the multi-tenant platform **102**. For example, the access tier **242** includes information about the products and/or services being offered by the service provider **108**, as well as service endpoints for accessing services (e.g., the service **116**) offered by the service provider **108**. The service endpoints can be referenced by a corresponding node in the hierarchical data structure **244** that represents the service provider **108**.

(34) Various embodiments of how the hierarchical data structure **244** is generated and accessed are discussed below with reference to FIGS. 3 and 4. The MT-IDP **246** can generate and manage representations of the entities of the onboarded tenants, i.e., entities **206(1)**, **206(2)**, **202(1)**, and **202(2)**, as well as the 3P-IDP **204** of the service provider **108**. Similarly, the MT-IDP **246** can generate and manage representations of entities **214(1)**, **214(2)**, **212** of the service provider **110**.

(35) Since each of the service providers **108** and **110** includes own 3P-IDP **204** and **212** respectively, each of these tenants can continue to manage the identities of their own respective entities. For example, the service provider **110** can continue to manage the entities **214(1)-214(2)**, including managing identity information, contact data, and any characteristics of the consumers **214(1)** and **214(2)**. The service provider **110** can update any changes to the entities **214(1)** and **214(2)** to the multi-tenant platform **102**, such as via a representation of the 3P-IDP **204** in the hierarchical data structure **244**. These 3P-IDP **212** originated updates are used to keep the data of their respective representations up-to-date. Similarly, any services (such as core services) performed via the representations of the hierarchical data structure **244** can be propagated back to

the 3P-IDP **212**.

(36) The transaction resource service **150** (referred to also as MT-FI **150**) can include a resource access tier **252** and a hierarchical data structure **254** that is managed by an FI-IDP **256**. The resource access tier **252** can customize resource data associated with entities of the service provider when being on-boarded with the multi-tenant platform **102**. The resource access tier **252** includes information about the resources **162** that is associated with each one of the entities of the respective service provider, such as resources associated with the consumer **202(1)** and/or resources associated with the consumer **214(1)**. The resource access tier can also include information about transaction services being offered by the service provider **108**, as well as service endpoints for accessing the transaction services that use the resources **162**. The service endpoints can be referenced by a corresponding node in the hierarchical data structure **254**. In some embodiments, the resources can be implemented as funding instruments, e.g., for the consumers **202(1)** and **202(2)**.

(37) Similarly, the social graph service **160** (referred to also as MT-SG **160**) can include a social access tier **262** and a hierarchical data structure **264** that is managed by an SG-IDP **266**. The social access tier **262** can customize social data associated with the entities of the service provider when being on-boarded with the multi-tenant platform **102**. The social access tier **262** includes information about links between each of the consumers, such as consumers **202(1)**, **202(2)**, **214(1)**, and/or **214(2)**. The social access tier **262** can also include service endpoints for accessing various social media applications based on which the hierarchical data structure **264** is generated.

(38) In some implementations, the multi-tenant platform **102** can enable co-ordination of multiple identity namespaces (e.g., separate namespaces associated with the service provider **108**, the service provider **110**, as well another namespace associated with the fully integrated service provider of the multi-tenant platform **102**) to efficiently and securely transfer transaction resources between identities corresponding to entities of these namespaces. For example, a user in one domain (e.g., a namespace associated with the service provider **108**) can, via the multi-tenant platform **102**, lookup users in other namespaces (e.g., a user in a namespace associated with the service provider **110**, or a merchant associated with the service provider **112**) and establish a relationship including transaction resource transfers.

(39) In some embodiments, a user (e.g., a customer) can onboard as an entity to any of the service providers **108-112**. In one example, the user can onboard onto the service provider **108**. As the service provider itself is a tenant in the multi-tenant architecture **102**, a representation of the user would be created by the identity services **132** (such as under a 3P-IDP representation for that tenant representation in the hierarchical data structure **244**). That 3P-IDP can be indicated as a system-of-record (SOR) for that onboarded user. Once the user has onboarded, the transaction resources (e.g., funding instruments) for the user would be managed by the MT-FI **150**. The user can add/remove transaction resources **162** by accessing the service provider **108**, which would be reflected by the MT-FI **150**. The multi-tenant platform **102** can identify the user in the namespace for the service provider **108**, and can also associate (via the identity services **132**) that user with the same user in another namespace (e.g., in the namespace for the service provider **110**). The multi-tenant platform **102** can use the identity services **132** to identify the same user and his/her transaction resources across various namespace, thus enabling the same user to have multiple accounts in different namespaces and potentially linking or federating these accounts.

(40) In some embodiments, during the onboarding process for a new tenant, there is a phase of tenant Setup where entities (e.g., consumer representations of the actual consumer, linked consumer for VENMO (e.g., the service provider **108**) consumer under VENMO tenancy are modeled with specific policies set for the tenant. When the tenant (e.g., VENMO) gets its credentials (client identifier and secret) from the multi-tenant platform **102**, the credentials can include appropriate Access Token and/or Refresh Tokens. These tokens can indicate embedded tenant information. When the tenant calls into the multi-tenant API **126**, the multi-tenant platform

102 can authorize the request and convert the Access Token into a Security token.

(41) When transaction resources are on-boarded onto the MT-FI **150**, the tenant identifier can be used by the MT-FI **150** and the associated details can be retrieved from the security context (of the security token). In some embodiments, some details around the tenant and the associated accounts can be inferred from the Access Token and the translated security context after validation/verifications from the identity services **132**. Thus, when a new Venmo user account is generated, the identity services **132** can create a representation of that account in the hierarchical data structure **244**, and associated funding instruments for that user can be added to the MT-FI **150**.

(42) FIG. 3 is a diagram illustrating embodiments of flow of operations of using the multi-tenant software architecture for unified transaction services. FIG. 3 illustrates a logical view of how transactions are processed between service providers **108** and **110** using the mt-transaction processor **101**. In some embodiments, any one of the service providers **108** or **110** can receive a request **302** to provide a transaction service to a user. The service provider **108** includes resources **162**. In some embodiments, at least a portion of the resources **162** (e.g., for certain users of the service provider **108**) can be associated with certain elements of the resources **164** at the service provider **110**. Thus, a transaction for a certain user that accesses the user's resources at the resources **162**, would instead use the associated element of the resources **164** at the service provider **164**.

(43) The resources **162** can be modeled by the MT-FI **150**, such as by using resource representations using the hierarchical data structure **254** that is managed by the FI-IDP **256**. In some embodiments, a transaction resource manager **306** can be used to manage resource representations that are mapped to each tenant represented by the MT-ID **132**. For example, the transaction resource manager **306** can be implemented as a separate wallet for the tenant/service provider **108**. The transaction resource manager **306** can manage separate MT-FIs for each customer of the tenant **108**.

(44) The mt-transaction processor **101** can process transactions, such as ones indicated by the request **302**. Transactions can be processed using the risk **138** and/or compliance services **140** based on identity data of the MT-ID **132** and resource data of the MT-FI **150**. Once the mt-transaction processor **101** determines the appropriate identity and resources for the transaction, they can be passed to the payment core service **148**. The payment core service **148** can process the transaction and use the external processing **312**, such as external processors, networks (e.g., credit card networks), and/or issuers.

(45) FIG. 4 is a diagram illustrating hierarchical data structures used by the multi-tenant software architecture systems that provide unified transaction services. FIG. 4 illustrates several hierarchical data structures **402**, **403**, and **404** (which collectively can be referred to as the hierarchical data structure **244**), which are accessible via the MT-IDP **246**. FIG. 4 illustrates the relationships between the entities of the hierarchical data structures **402-404**. The organization and linking of the hierarchical data structures of FIG. 4 can be referred to as a dependency graph. It is noted that the organization and type of the hierarchical data structures **402-404** is shown for illustrative purposes only, and that one or more of 3P-IDP representations **406** and **411** can implement different data structures, as desired.

(46) The first hierarchical data structure **402** illustrates service providers **409**, **412(1)**, and **412(2)** that are fully integrated, and thus offer core services to a merchant **418**, consumer **416**, lightweight merchant **420(1)**, as well as to entities of an on-boarded service provider that is shown by second hierarchical data structure **404**. In FIG. 4, the entities **412(1)**, **412(2)**, **414(1)**, **414(2)**, **416**, and **420(1)** are fully integrated into the multi-tenant platform **102**, and are managed by the MT-IDP **246**.

(47) The hierarchical data structure **404** can correspond to the on-boarded service provider **110** and is managed by the 3P-IDP representation **406**. Thus, the 3P-IDP representation **406** can be a representation, in the hierarchical data structure **244**, of the actual 3P-IDP **212** used by the service

provider **110**. The Identity-as-a-Service (IaaS) services can further map different schema used by the 3P-IDP **212** to the hierarchical data structure **404**. During onboarding of the service provider **110**, the multi-tenant platform **102** can generate representations **408(1)** and **408(2)** for the consumers **214(1)** and **214(2)** that are managed by that tenant's 3P-IDP **212**. The consumers **408(1)** and **408(2)** are not fully integrated into the multi-tenant platform **102** (e.g., unlike consumer **416**), and are instead on-boarded onto the multi-tenant platform **102**. The multi-tenant platform **102** can further generate representations in the first hierarchical data structure **402** by generating linked consumers **414(1)** and **414(2)**. A linked consumer is an entity that represents a corresponding entity in another hierarchical data structure, and that contains a limited amount of the data for that corresponding entity. The linked consumers **414(1)** and **414(2)** link to the consumers **408(1)** and **408(2)**, respectively.

(48) During the process of onboarding the service provider **108**, the multi-tenant platform **102** can similarly generate the hierarchical data structure **403** that is managed by the 3P-IDP representation **411**. As shown, the multi-tenant platform **102** can generate representations **432(1)** and **432(2)** of the merchants **206(1)** and **206(2)**, respectively. The actual identities of the merchants **206(1)** and **206(2)** are managed by that tenant's 3P-IDP **204**, and their representations **432(1)** and **432(2)** are managed by a 3P-IDP representation **411** of the tenant's 3P-IDP **204**. The merchants **432(1)** and **432(2)** are not fully integrated into the multi-tenant platform **102** (unlike merchant **418**), and are instead on-boarded onto the multi-tenant platform **102**. During the on-boarding process, the multi-tenant platform **102** can also generate a lightweight merchant **420(1)** that links to the merchants **432(1)**. A lightweight merchant is an entity that represents a corresponding entity in another hierarchical data structure, and that contains some of the data for that corresponding entity.

(49) The hierarchical data structure **402** indicates that the linked consumers **414(1)** and **414(2)** are child nodes of the node of the service provider **412(1)**, and thus are acted upon by the service provider **412(1)**. However, the service provider **412(1)** does not directly modify the linked to consumers **408(1)** and **408(2)**. Similarly, the hierarchical data structure **402** indicates that the lightweight merchant **420(1)** is a child node of the node of the service provider **412(2)**, and thus are acted upon by the service provider **412(2)**. However, the service provider **412(2)** does not directly modify the linked to merchant **432(1)**. An example of this is where the merchant **432(1)** is operating as merchant of a service provider indicated by the 3P-IDP representation **409**, which can be a marketplace tenant. Although the lightweight merchants **420(1)** can inherit any access permissions from the service provider **412(2)**, the 3P-IDP representation **411** provides the access to the merchant **206(1)** that is represented by **432(1)**. In case of access permission conflict, access permissions indicated by merchants **206(1)** prevails, followed by access permissions indicated by merchant **432(1)**, and lastly by the lightweight merchant **420(1)**. However, in some cases, the multi-tenant architecture **102** can be configured to change and/or propagate any access permissions of the merchant **432(1)** and/or merchant **206(1)** based on access permissions of the lightweight merchant **420(1)**.

(50) FIG. 5 is a diagram illustrating embodiments of relationships between a multi-tenant identity service, multi-tenant transaction resource service, and multi-tenant social graph service for unified transaction services. FIG. 5 illustrates example relationships between various data structures and services used by the multi-tenant platform **102** for a particular user. FIG. 5 is described with reference to the examples shown in FIGS. 1-4 for ease of explanation.

(51) The hierarchical data structure **244** manages the multi-tenant identity and includes the MT-IDP **246** that links to a service provider **412(1)**, which in turn is linked to two linked consumers **414(1)** and **414(2)**. The hierarchical data structure **244** can include similar content to what is shown by FIG. 4, with simplifications for readability purposes. Various implementations of generating the hierarchical data structure **244** when onboarding tenants are discussed above with reference to FIGS. 1-4.

(52) The hierarchical data structure **264** includes the SG-IDP **266** that manages the social graph

connections for each user, includes the node for a user **501**, as well as a connection graph **510** that indicates the social connections between other users and the user **501**. The connection graph **510** shows links between various users and the order of their associations. The MT-SG domain supports a network model of users associated with tenants, where these users can be associated with MT-IDP linked consumers (i.e., that correspond to various users). Each of these users, including the user **501**, can be associated with a unique resource representation (e.g., as discussed with reference to the transaction resource service). The LC **414(1)** can include a user ID that corresponds to the user **501** element in the hierarchical data structure **264**. The LC **414(1)** can also include a resource representation that corresponds to the resource representation **502** in the hierarchical data structure **254**.

(53) The hierarchical data structure **254** includes the FI-IDP **256** that manages the resources **504-510**, which can represent a portion of the resources **162** that are associated with the user. As noted above, this portion of the resources **162** can be mapped to the resources **164**. This mapping can be noted by the resource representation **502**. Thus, the MT-FI can model a tenant-based transaction resource manager that is associated with the consumer's representation and transaction resources. The transaction resource manager can include the associated transaction resources **504-510**, which for implementation of financial instruments, can correspond to a bank, account balance, and/or types of cards (Credit, Debit, Pre-paid for the user). As shown in the diagram, the resource representation **502** can be associated with a particular tenant (i.e., the service provider **412(1)**) that is associated with various transaction resources including **504-510**. In some implementations, the service provider **412(1)** may support multiple resource representations.

(54) The MT-SG **160** allows for managing social relationships between identities across different namespaces. As shown in FIG. 5, the user **501** can be linked, via the LC **414(1)**, from the hierarchical data structure **244** of the identity services **132**. The hierarchical data structure **264** associated with the user **501** can indicate that the user **501** interacts with various users represented by nodes in the graph portion **510**. These nodes, that can represent identities of other users in the namespace of the tenant associated with the service provider **412(1)**, can be derived from the MT-IDP **246**. The links between the nodes of the graph portion **510** can represent various characteristics of interaction between the users. For transaction services, the interaction can be characterized with a transaction amount, a type of transaction resources used, an actual source of the transaction resource. Other such characteristics can indicate any messages exchanges, similar interests between the users, co-location, proximity, relationships (friend, spouse, family, etc.) and/or other attributes that qualify these links. Once the hierarchical data structure **244** is established, for any future interactions of the user **501**, the hierarchical data structure of the MT-SG **160** can be used to build user experiences. For example, for sending actual transaction resources from the user's **501** account, the mt-transaction processor **101** can determine certain relevant receivers based on the MT-SG **160**.

(55) FIG. 6 is a flow diagram illustrating embodiments of operations for using multi-tenant software architecture systems that provide unified transaction services. The method of FIG. 6 is described with reference to the systems and components described in FIGS. 1-5 (for illustration purposes and not as a limitation). The example operations can be carried out by the transaction processor that uses the multi-tenant platform **102**.

(56) Beginning with **602**, the processing system **100** receives a request at a first provider to provide a service to a user. The first provider can be a service provider **208** that is on-boarded by the multi-tenant platform **102**, and thus can provide on-boarded services to the users via the multi-tenant platform **102**. In some embodiments, the request can be received by the actual service provider, such as by the service provider **108** (as illustrated by FIG. 3). The request can be for a transaction service that is provided by the onboarded service provider. The transaction service can be for transaction resources that appear to be a part of the service provider **108**, but are actually hosted by another service provider, such as by the service provider **164**. In some embodiments, the multi-

tenant platform **102** can receive the request from the user device **104** for a transaction that appears to be provided by the multi-tenant platform **102** and that uses resources that appear to be hosted by the multi-tenant platform **102**. The actual service that provides the requested transaction can be performed by another certain service on the multi-tenant platform **102** for a fully on-boarded service provider, or can be performed by another service at another service provider that isn't fully on-boarded.

(57) At **604**, the multi-tenant platform **102** accesses a representation of the first service provider in a first hierarchical data structure that is managed by the second service provider. With reference to FIG. 4, the multi-tenant platform **102** can access the hierarchical data structure **403** that represents the actual service provider **108**, where the hierarchical data structure **403** is managed by the MT-IDP **246**. With reference to FIG. 5, the multi-tenant platform **102** can access the LC **414(1)** in the hierarchical data structure **244** that is managed via the MT-IDP **246**. The multi-tenant platform **102** can also access the LC **506** in the hierarchical data structure **254** that is managed via the FI-IDP **256**.

(58) At **606**, the multi-tenant platform **102** determines whether the transaction resource for the transaction is available at the second service provider using a resource representation. At **606**, the multi-tenant platform **102** can first determine the transaction resource needed for the requested transaction. The multi-tenant platform **102** can determine whether the transaction resources corresponding to the resource representation are accessible via the first service provider or the second service provider. The multi-tenant platform **102** can determine that the transaction resources required for completion of the first transaction service are accessible at the second service provider based on the request which can indicate one or more of a type and a source of the transaction resources. The multi-tenant platform **102** can determine whether a second hierarchical data structure, that is managed by the first transaction service, includes one or more transaction resources that are linked with the first representation in the first hierarchical data structure. The multi-tenant platform **102** can, responsive to determining that the second hierarchical data structure does not include the one or more transaction resources that are linked with the first representation, generate the resource representation at the second resource provider without linking to the first resource provider. If the transaction resource is available at the second service provider, flow continues to **610**. Otherwise, flow continues to **608**.

(59) At **608**, the multi-tenant platform **102** determines which service provider provides the transaction resource. For example, the determination is for whether the transaction resource is one of the core services **130(1)** available at the multi-tenant platform **102** (acting as the first service provider), or is a service that is provided by another service provider (also referred to as a "requested service provider").

(60) At **610**, the multi-tenant platform **102** determines whether the transaction resource is accessible at the provided service provider via the resource representation. The multi-tenant platform **102** can determine that the required transaction resources are accessible via the resource representation based on determining that a second hierarchical data structure, that is managed by the first transaction service, includes the one or more transaction resources that are linked with the first representation in the first hierarchical data structure. For example, the multi-tenant platform **102** can determine that additional user information (for the user and/or the second service provider) that is needed to access the service is not available at the first hierarchical data structure. For example, the multi-tenant platform **102** can determine certain user data from the representation in the first data structure, but also determine that additional user information is needed to perform the service. If the user data is available at another service provider, flow continues to **612**. Otherwise, flow continues to **616**.

(61) At **612**, the multi-tenant platform **102** determines a service provider through which the transaction resource can be accessed via the resource representation at the provided service provider. For example, with reference to FIG. 3, the mt-transaction processor **101** can determine

that the resources **164** (at the service provider **110**) can be accessed via the resources **162** (at the service provider **108**). In some embodiments, the multi-tenant platform **102** can determine to use another transaction processor instead of the mt-transaction processor **101**. For example, the multi-tenant platform **102** can determine to use another transaction processor of the service provider **110** for processing (including accessing of the resource representation discussed below). The other transaction processor can be modelled as a representation transaction processor by the multi-tenant platform to facilitate access of the actual transaction processor at the service provider **110**.

(62) At **616**, the multi-tenant platform **102** accesses, at the appropriate service provider, the transaction resource using the resource representation. The multi-tenant platform **102** can use the mt-transaction processor **101** to access the transaction resources **164** at the first service provider **110** via the resource representation. The mt-transaction processor **101** can be configured to route the first transaction service with reference to the transaction resource **164** through additional services, such as services **138**, **140**, and/or **312**. In some embodiments, as discussed above, the multi-tenant platform **102** can use, via the representation of the service provider, the actual transaction processor of the service provider **110** to access the transaction resources **164** at the first service provider **110** via the resource representation.

(63) The multi-tenant platform **102** can determine a user representation of the user at third service provider, the third service provider modeling a social graph of connections between the user and a plurality of users. The multi-tenant platform **102** can process the first transaction service using the social graph. The third service provider can be accessible, at the first service provider, via the second service provider using the first representation. The multi-tenant platform **102** can generate a resource request for consuming a first amount of transaction resources, via the resource representation, for the first transaction service. With reference to FIG. **3**, the transaction processor **101** can generate a resource request to access the transaction resource **164** at the service provider **110**.

(64) It should be understood that FIGS. **1-6** and the operations described herein are examples meant to aid in understanding embodiments and should not be used to limit embodiments or limit scope of the claims. Embodiments may perform additional operations, fewer operations, operations in a different order, operations in parallel, and some operations differently. For example, one or more elements, steps, or processes described with reference to the flow diagrams of FIGS. **5** and **6** may be omitted, described in a different sequence, or combined as desired or appropriate.

(65) As will be appreciated by one skilled in the art, aspects of the present disclosure may be embodied as a system, method, or computer program product. Accordingly, aspects of the present disclosure may take the form of an entirely hardware embodiment, a software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a “module” or “system.” Furthermore, aspects of the present disclosure may take the form of a computer program product embodied in one or more computer readable medium(s) having computer readable program code embodied thereon.

(66) Any combination of one or more computer readable medium(s) may be utilized. The computer readable medium may be a computer readable signal medium or a computer readable storage medium. A computer readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. More specific examples (a non-exhaustive list) of the computer readable storage medium would include the following: a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a portable compact disc read-only memory (CD-ROM), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer readable storage medium may be any tangible and/or non-transitory medium that can contain, or store a program for use by

or in connection with an instruction execution system, apparatus, or device.

(67) A computer readable signal medium may include a propagated data signal with computer readable program code embodied therein, for example, in baseband or as part of a carrier wave. Such a propagated signal may take any of a variety of forms, including, but not limited to, electromagnetic, optical, or any suitable combination thereof. A computer readable signal medium may be any computer readable medium that is not a computer readable storage medium and that can communicate, propagate, or transport a program for use by or in connection with an instruction execution system, apparatus, or device.

(68) Computer program code embodied on a computer readable medium may be transmitted using any appropriate medium, including but not limited to wireless, wireline, optical fiber cable, RF, etc., or any suitable combination of the foregoing.

(69) Computer program code for carrying out operations for aspects of the present disclosure may be written in any combination of one or more programming languages, including an object-oriented programming language such as Java, Smalltalk, C++ or the like and conventional procedural programming languages, such as the “C” programming language or similar programming languages. The computer program code may execute (e.g., as compiled into computer program instructions) entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider).

(70) Aspects of the present disclosure are described with reference to flow diagram illustrations and/or block diagrams of methods, apparatus (systems) and computer program products according to embodiments of the present disclosure. It will be understood that each block of the flow diagram illustrations and/or block diagrams, and combinations of blocks in the flow diagram illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the computer program instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the flow diagrams and/or block diagram block or blocks.

(71) These computer program instructions may also be stored in a computer readable medium that can direct a computer, other programmable data processing apparatus, or other devices to function in a particular manner, such that the instructions stored in the computer readable medium produce an article of manufacture including instructions which implement the function/act specified in the flow diagram and/or block diagram block or blocks.

(72) The computer program instructions may also be loaded onto a computer, other programmable data processing apparatus, or other devices to cause a series of operational steps to be performed on the computer, other programmable apparatus or other devices to produce a computer implemented process such that the instructions which execute on the computer or other programmable apparatus provide processes for implementing the functions/acts specified in the flow diagrams and/or block diagram block or blocks.

(73) FIG. 7 is a block diagram of one embodiment of an electronic device **700** used in the communication systems of FIGS. 1-6. In some implementations, the electronic device **700** may be a laptop computer, a tablet computer, a mobile phone, a kiosk, a powerline communication device, a smart appliance (PDA), a server, and/or one or more other electronic systems. For example, a user device may be implemented using a mobile device, such as a mobile phone or a tablet computer. For example, a payment system may be implemented using one or more servers. The electronic device **700** can include a processor unit **702** (possibly including multiple processors,

multiple cores, multiple nodes, and/or implementing multi-threading, etc.). The electronic device **700** can also include memory unit **706** and may access machine medium **712**. The memory unit **706** may be system memory (e.g., one or more of cache, SRAM, DRAM, zero capacitor RAM, Twin Transistor RAM, eDRAM, EDO RAM, DDR RAM, EEPROM, NRAM, RRAM, SONOS, PRAM, etc.) or any one or more of the above already described possible realizations of machine-readable media (such as shown by machine medium **712**). The electronic device **700** can also include a bus **710** (e.g., PCI, ISA, PCI-Express, HyperTransport®, InfiniBand®, NuBus, AHB, AXI, etc.), and network interfaces **704** can include wire-based interfaces (e.g., an Ethernet interface, a powerline communication interface, etc.). The electronic device **700** includes a communication interface **708** for network communications. The communication interface **708** can include at least one of a wireless network interface (e.g., a WLAN interface, a Bluetooth interface, a WiMAX interface, a ZigBee interface, a Wireless USB interface, etc.). In some implementations, the electronic device **700** may support multiple network interfaces—each of which is configured to couple the electronic device **700** to a different communication network.

(74) The memory unit **706** can embody functionality to implement embodiments described in FIGS. **1-6** above. In one embodiment, the memory unit **706** can include one or more of functionalities for using unified transaction services in a multi-tenant architecture system. Any one of these functionalities may be partially (or entirely) implemented in hardware and/or on the processor unit **702**. For example, some functionality may be implemented with an application specific integrated circuit, in logic implemented in the processor unit **702**, in a co-processor on a peripheral device or card, etc. Further, realizations may include fewer or additional components not illustrated in FIG. **7** (e.g., video cards, audio cards, additional network interfaces, peripheral devices, etc.). The processor unit **702**, memory unit **706**, the network interfaces **704**, and the communication interface **708** are coupled to the bus **710**. Although illustrated as being coupled to the bus **710**, the memory unit **706** may be coupled to the processor unit **702**.

(75) While the embodiments are described with reference to various implementations and exploitations, it will be understood that these embodiments are illustrative and that the scope of the present disclosure is not limited to them. In general, techniques for using unified transaction services in a multi-tenant architecture system as described herein may be implemented with facilities consistent with any hardware system or hardware systems. Many variations, modifications, additions, and improvements are possible.

(76) Plural instances may be provided for components, operations or structures described herein as a single instance. Finally, boundaries between various components, operations and data stores are somewhat arbitrary, and particular operations are illustrated in the context of specific illustrative configurations. Other allocations of functionality are envisioned and may fall within the scope of the present disclosure. In general, structures and functionality presented as separate components in the exemplary configurations may be implemented as a combined structure or component. Similarly, structures and functionality presented as a single component may be implemented as separate components. These and other variations, modifications, additions, and improvements may fall within the scope of the present disclosure.

Claims

1. A method for using unified identity services in a multi-tenant architecture system, the method comprising: accessing a request, associated with a first service provider, to provide a first transaction service for an entity registered with the first service provider, the first service provider configured to provide a first set of services for the entity; accessing a first representation of the first service provider in a first hierarchical data structure, the first hierarchical data structure being managed by a second service provider, the second service provider managing a plurality of user identities of a plurality of users; determining, based on the first representation, that a first

transaction resource required for processing of the first transaction service is provided at a third service provider and accessible via a resource representation, wherein the resource representation is implemented as part of a second hierarchical data structure and provides access to a plurality of transaction resources including the first transaction resource, and wherein at least one of the first hierarchical data structure or the second hierarchical data structure contains a plurality of nodes that each represent a different user account of a different user of the plurality of users; responsive to determining that the first transaction resource is provided at the third service provider, accessing, at the second service provider, the first transaction resource via the resource representation; determining a user representation of a user of the plurality of users at the third service provider based at least in part on a social graph comprising connections between a node of the plurality of the nodes that represent the user and a subset of the plurality of nodes that represent other users of the plurality of users, wherein the connections of the social graph represent characteristics of interactions between the user and the other users, and wherein the characteristics comprise one or more of: a transaction amount, a type of transaction resource used, an actual source of the transaction resource, a message exchange between two or more of the plurality of users, a common interest between two or more of the plurality of users, a common location between two or more of the plurality of users, or a proximity between two or more of the plurality of users; and processing the first transaction service using the social graph.

2. The method of claim 1, wherein the first service provider is configured to provide, via a network, a first set of services to one or more devices associated with the entity, wherein the first set of services does not include functionality of the first transaction service.

3. The method of claim 1, further comprising accessing a transaction processor using the first transaction resource at the second service provider via the resource representation, wherein the transaction processor is configured to route the first transaction service with reference to the first transaction resource through additional services.

4. The method of claim 1, further comprising generating a resource request for consuming a first amount of the first transaction resource, via the resource representation, for the first transaction service.

5. The method of claim 1, further comprising onboarding the third service provider onto the first hierarchical data structure of the second service provider to expose services provided by the third service provider to a linked identity reference from the second service provider.

6. The method of claim 1, wherein the second hierarchical data structure is linked with the first hierarchical data structure.

7. The method of claim 1, wherein the second hierarchical data structure provides access to the plurality of transaction resources to a plurality of entities represented by the first hierarchical data structure.

8. A system comprising: a non-transitory memory storing instructions; and a processor configured to execute the instructions to cause the system to: receive a request, from a first service provider, to provide a first transaction service for a user registered with the first service provider; access a first representation of the first service provider in a first hierarchical data structure, the first hierarchical data structure being managed by a second service provider, the second service provider managing a plurality of user identities of a plurality of users that include the user; determine, based on the first representation, that a first transaction resource required for processing of the first transaction service is provided at a third service provider and accessible via a resource representation associated with a second hierarchical data structure, wherein the second hierarchical data structure provides access to a plurality of transaction resources including the first transaction resource, and wherein the first hierarchical data structure and the second hierarchical data structure each contains a plurality of nodes that represent a plurality of different user accounts, respectively; responsive to determining that the first transaction resource is provided at the third service provider, access, at the second service provider, the first transaction resource via the resource representation; determine, at

least in part via a social graph, a user representation of a user of the plurality of users at the third service provider, wherein the social graph is modeled by the third service provider and comprises a plurality of connections interconnecting a plurality of nodes, wherein each node of the plurality of nodes represents a respective user of the plurality of users, and wherein each connection of the plurality of connections corresponds to a characteristic of an interaction between the nodes interconnected by the connection; and processing the first transaction service using the social graph.

9. The system of claim 8, wherein executing the instructions further cause the system to, access a transaction processor using the first transaction resource at the second service provider via the resource representation, wherein the transaction processor is configured to route the first transaction service with reference to the first transaction resource through additional services.

10. The system of claim 8, wherein executing the instructions further cause the system to, generate a resource request for consuming a first amount of the first transaction resource, via the resource representation, for the first transaction service.

11. The system of claim 8, wherein executing the instructions further cause the system to, onboard the third service provider onto the first hierarchical data structure of the second service provider to expose services provided by the third service provider to a linked identity reference from the second service provider.

12. The system of claim 8, wherein the second hierarchical data structure provides access to the plurality of transaction resources to a plurality of entities represented by the first hierarchical data structure.

13. A non-transitory machine-readable medium having instructions stored thereon, the instructions executable to cause performance of operations comprising: receiving a request, from a first service provider, to provide a first transaction service for a user of a plurality of users registered with the first service provider; accessing a first representation of the first service provider in a first hierarchical data structure, the first hierarchical data structure being managed by a second service provider, the second service provider managing a user identity of the user; determining, based on the first representation, that a first transaction resource required for processing of the first transaction service is provided at a third service provider and accessible via a resource representation that is implementable as a part of a second hierarchical data structure, wherein the second hierarchical data structure comprises a different set of nodes than the first hierarchical data structure, the nodes representing different user accounts, wherein the second hierarchical data structure enables an access to at least the first transaction resource; responsive to determining that the first transaction resource is provided at the third service provider, accessing, at the second service provider, the first transaction resource via the resource representation; accessing a user representation of the user at the third service provider, wherein the user representation is associated with a social graph comprising connections between a node of the plurality of the nodes that represent the user and a subset of the plurality of nodes that represent other users of the plurality of users, wherein the connections of the social graph represent characteristics of interactions between the user and the other users, and wherein the characteristics comprise one or more of: a transaction amount, a type of transaction resource used, an actual source of the transaction resource, a message exchange between two or more of the plurality of users, a common interest between two or more of the plurality of users, a common location between two or more of the plurality of users, or a proximity between two or more of the plurality of users; and processing the first transaction service using the social graph.

14. The non-transitory machine-readable medium of claim 13, wherein the operations further comprise: accessing a transaction processor using the first transaction resource at the second service provider via the resource representation, wherein the transaction processor is configured to route the first transaction service with reference to the first transaction resource through additional services.

15. The non-transitory machine-readable medium of claim 13, wherein the operations further comprise: generating a resource request for consuming a first amount of the first transaction resource, via the resource representation, for the first transaction service.
 16. The non-transitory machine-readable medium of claim 13, wherein the operations further comprise: onboarding the third service provider onto the first hierarchical data structure of the second service provider to expose services provided by the third service provider to a linked identity reference from the second service provider.
 17. The non-transitory machine-readable medium of claim 13, wherein the second hierarchical data structure provides access to a plurality of transaction resources, including the transaction resource, to a plurality of entities represented by the first hierarchical data structure.
 18. The method of claim 1, wherein the social graph is provided by a multi-tenant graph service that includes service endpoints that access a plurality of social media applications.
 19. The system of claim 8, wherein the characteristic comprises: a transaction amount, a type of transaction resource used, an actual source of the transaction resource, a message exchange between two or more of the plurality of users, a common interest between two or more of the plurality of users, a common location between two or more of the plurality of users, or a physical proximity between two or more of the plurality of users, and wherein the social graph is provided by a multi-tenant graph service that includes service endpoints that enable accessing a plurality of social media applications.
 20. The non-transitory machine-readable medium of claim 13, wherein the social graph is provided by a multi-tenant graph service that includes service endpoints that enable accessing a plurality of social media applications.
-